

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456109.9012 +/- 0.0042 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.138 +/- 0.042
Transit depth (Rp/Rs)^2	1.91 +/- 1.15 [%]
Orbital Inclination (inc)	85.42 +/- 1.44
Airmass coefficient 1 (a1)	6.87 +/- 3.18
Airmass coefficient 2 (a2)	-1.11 +/- 0.16
Scatter in the residuals of the lightcurve fit is	77.74 %
Best Comparison Star	#2 - [274, 140]
Optimal Aperture	5.95
Optimal Annulus	16.92
Transit Duration (day)	0.1287 +/- 0.0047

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.138 $\pm$ 0.042 vs 0.1036 (deviation 0.57 σ)
Duration fit vs published	0.1287 d vs 0.1567 d (deviation 4.17 σ)
Mid-time O-C	-10.0 min
Depth SNR	2.3
Residual scatter	77.74 %

Light curve

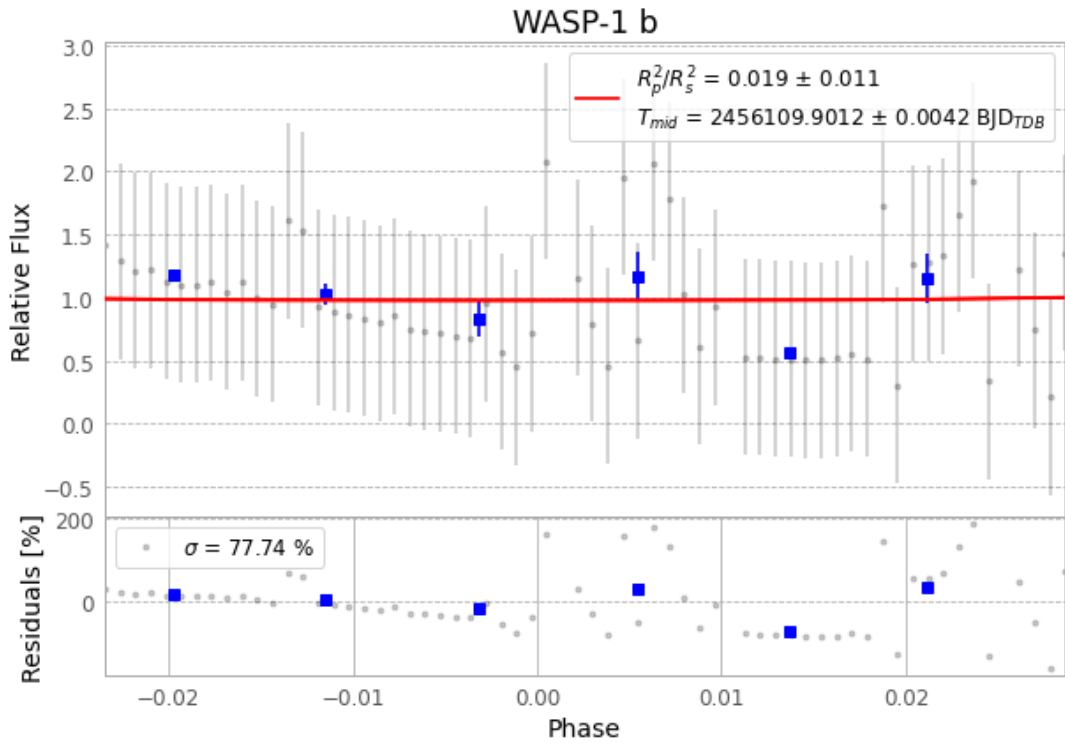


Figure 1 — FinalLightCurve_WASP-1 b_2012-07-01.png (SHA-256 ec931c218f7f0ead...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [267, 179], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 44b514a9034b0227...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

64 FITS frames (42 MB), WASP-1120701081212.FITS → WASP-1120701112113.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-1-120701.log (d21cf072b69a...), FinalLightCurve_WASP-1 b_2012-07-01.png (ec931c218f7f...), FinalLightCurve_WASP-1 b_2012-07-01.csv (d98bdcf41a1c...)

Compute resources

Wall time 195.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-17 23:20Z.